The interpolation law for the uniform doubling condition in general position

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Abstract

The companion paper [7] determined the critical scaling of the uniform doubling condition (UDC) of [1] for the orthogonal model $B^n \cap \{x_1 \le 0, \dots, x_k \le 0\}$, where a round ball is cut by mutually orthogonal half-spaces through the origin. Here we remove both idealizations. We treat an admissible configuration: a C^2 strictly convex body K with principal curvatures in a fixed band $[\lambda_{\min}, \lambda_{\max}]$, truncated by k half-spaces $\langle x, a_i \rangle \le c_i$ in general position, meaning that at each point p of the boundary stratum $E = \partial K \cap \bigcap_i \{\langle x, a_i \rangle = c_i\}$ the outer normal $\nu_K(p)$ and the truncating normals $a_1, \dots, a_k$ are linearly independent with pairwise angles in $[\theta_0, \pi - \theta_0]$. The stratum E is then a j-manifold with $j = n - 1 - k$. We prove that the critical scaling factor is

$$\ell_N \asymp \Big(\frac{\log N}{N}\Big)^{2/(2n-j)} = \Big(\frac{\log N}{N}\Big)^{2/(n+1+k)},$$

exponent and logarithm exact, with constants depending only on n, k, θ_0 and the curvature band. The mechanism that replaces the orthogonal model's axis-parallel windows and its exact reflection fraction 2^{-k} is the *conjugate frame*: the dual basis $f_1, \dots, f_k$ to the truncating normals inside their span, chosen tangent to K at p, in which the k constraints become the diagonal slab $\{s_i \le 0\}$ and the cap exponent $(n+1+k)/2$ is exposed as angle-independent. The reflection fraction is replaced by the normalized solid angle ω of the normal cone $N(G, p) = \mathrm{cone}\{\nu_K(p), a_1, \dots, a_k\}$, which is positive precisely because the configuration is in general position. We note that the natural attempt to reduce to the orthogonal model by a John-position rounding is provably blocked — no single linear map can keep B^n spherical and orthogonalize non-orthogonal normals — so the conjugate-frame argument is intrinsic by necessity. This completes the flat-transversal part of the program of [7, 5, 6]. The curved companion [8] proves the same exponent law for each fixed curved truncation family and isolates the remaining curvature-band-uniform lower/slab constant problem.

1 Setting and main results

1.1 The configuration

Let $n \ge 2$. Let $K \subset \mathbb{R}^n$ be a compact convex body with C^2 boundary whose principal curvatures lie in a fixed band $0 < \lambda_{\min} \le \lambda_{\max} < \infty$; in particular K is strictly convex, the Gauss map $\nu_K : \partial K \to S^{n-1}$ is a diffeomorphism, and at each $p \in \partial K$ the second fundamental form II_p satisfies $\lambda_{\min} |v|^2 \le \mathrm{II}_p(v, v) \le \lambda_{\max} |v|^2$ for tangent v. Let μ be the normalized Lebesgue measure on K.

Fix unit vectors $a_1, \dots, a_k \in S^{n-1}$ and reals $c_1, \dots, c_k$ with $1 \le k \le n-1$ (the case $k = 0$ is the smooth body, treated in [3]), and set the *truncation region*

$$G \;:=\; K \cap \bigcap_{i=1}^{k} \{x : \; \langle x, a_i \rangle \le c_i\}.$$

We call a_i the truncating normals and $h_k := \{(a_1, c_1), \dots, (a_k, c_k)\}$ the truncation set. The boundary of G is stratified by the active constraint set: for $S \subseteq \{1, \dots, k\}$ the stratum is $\partial K \cap \bigcap_{i \in S} \{\langle x, a_i \rangle = c_i\} \cap \bigcap_{i \notin S} \{\langle x, a_i \rangle < c_i\}$ together with the curved face $\{x \in \operatorname{int} K : \; \langle x, a_i \rangle = c_i \; (i \in S)\}$. The lowest transversal stratum is

$$E \;:=\; \partial K \cap \bigcap_{i=1}^{k} \{x : \; \langle x, a_i \rangle = c_i\}.$$

Definition 1.1 (Admissible / general position)**.** The configuration (K, h_k) is *admissible*, or the truncation is in *general position*, if there is $\theta_0 \in (0, \pi/2)$ such that $E \neq \emptyset$ and at every point $p \in E$:

(GP1) the active normals $\nu_K(p), a_1, \dots, a_k$ are linearly independent;

(GP2) all pairwise angles among $\nu_K(p), a_1, \dots, a_k$ lie in $[\theta_0, \pi - \theta_0]$; equivalently, the Gram matrix of the unit active normals has smallest eigenvalue $\ge g_0 = g_0(\theta_0, k) > 0$.

Under (GP1)–(GP2) the map $x \mapsto (\langle x, a_1 \rangle, \dots, \langle x, a_k \rangle)$ is a submersion of ∂K near E, so E is a compact C^2 submanifold of ∂K of dimension

$$j \;:=\; n - 1 - k \;\ge\; 0.$$

Throughout we keep $0 \le j \le n-1$; for $j = 0$ (i.e. $k = n-1$) the stratum E is a finite set of conical points and the exponent is the conical $1/n$ of [3, 4], so the interesting range is $1 \le j \le n-1$, i.e. $1 \le k \le n-2$. The orthogonal model of [7] is the special case $K = B^n$, $a_i = e_i$, $c_i = 0$, for which $E = \{x \in S^{n-1} : x_1 = \cdots = x_k = 0\}$, $\nu_K(p) = p$, all active normals orthogonal, and any $\theta_0 \le \pi/2$ admissible.

We assume, as in all the companion papers, that G has nonempty interior and is uniformly thick at scale $r_0 = r_0(K, h_k, \theta_0) > 0$: there is a ball $B(c^-, r_0) \subseteq G$. Such a ball exists for any admissible configuration because E is a transversal intersection, so G contains a fixed cone-cap neighborhood pointing into $\bigcap_i \{\langle x, a_i \rangle < c_i\}$; we record this as a standing hypothesis and absorb r_0 into the constants.

1.2 The conjugate frame

Fix $p \in E$. The truncating normals $a_1, \dots, a_k$ are linearly independent and, by (GP1), independent from $\nu_K(p)$. We will need to write the k constraints in coordinates that diagonalize them while respecting the tangent geometry at p.

Definition 1.2 (Conjugate frame)**.** Let $V := \operatorname{span}\{a_1, \dots, a_k\}$ and let $\Pi : \mathbb{R}^n \to T_p \partial K = \nu_K(p)^{\perp}$ be the orthogonal projection onto the tangent space at p. Because $\nu_K(p) \notin V$ by (GP1), $\Pi|_V : V \to \Pi V$ is injective, so $\bar{a}_i := \Pi a_i$ are linearly independent tangent vectors. The *conjugate frame* $f_1, \dots, f_k$ is the basis of $\Pi V \subseteq T_p \partial K$ dual to $(\bar{a}_1, \dots, \bar{a}_k)$:

$$\langle f_i, a_j \rangle = \delta_{ij}, \qquad \langle f_i, \nu_K(p) \rangle = 0 \qquad (1 \le i, j \le k).$$

The two displayed conditions determine $f_1, \dots, f_k$ uniquely: they force $f_i \in T_p\partial K$, and on $T_p\partial K$ the functional $v \mapsto \langle v, a_j\rangle = \langle v, \bar{a}_j\rangle$ has the matrix $\bar{G} = (\langle \bar{a}_i, \bar{a}_j\rangle)$, so $f_i = \sum_m (\bar{G}^{-1})_{im}\bar{a}_m$. By (GP2) the Gram matrix of a_i has eigenvalues in $[g_0, k]$, and the tangential projection along $\nu_K(p)$ changes lengths by a factor bounded in terms of θ_0 only, so

$$g_1(\theta_0, k) \;\le\; |f_i| \;\le\; g_2(\theta_0, k), \qquad |\langle f_i, f_j\rangle| \le g_2^2, \tag{1}$$

with $g_1 > 0$. In the new linear coordinates

$$s_i \;:=\; \langle x - p, a_i\rangle \qquad (1 \le i \le k),$$

the truncation constraints $\langle x, a_i\rangle \le c_i$ read exactly $s_i \le 0$, since $\langle p, a_i\rangle = c_i$ for $p \in E$. Thus the conjugate frame diagonalizes the truncation: the k half-spaces become the standard orthant $\{s_1 \le 0, \dots, s_k \le 0\}$, and f_i is the direction in which only s_i moves (to first order, tangentially). This is the device that replaces the relation $\langle w, \tilde{u}\rangle = 0$ used in the orthogonal model: there $a_i = e_i$ is self-dual, $f_i = e_i$, and the windows are axis-parallel; in general position the windows are spanned by the conjugate frame instead.

The invariant content we use repeatedly is the splitting of $T_p\partial K$ into the k-dimensional conjugate span ΠV and its j-dimensional orthogonal complement

$$T_pE \;=\; \{v \in T_p\partial K :\; \langle v, a_i\rangle = 0 \;\; \forall i\}, \qquad \dim T_pE = m := j = n - 1 - k,$$

the tangent space to the stratum; concrete orthonormal coordinates on each factor are introduced as needed in §2.1.

1.3 Main results

The first result isolates the exponent. For $p \in E$ let

$$N(G, p) \;:=\; \operatorname{cone}\{\nu_K(p), a_1, \dots, a_k\}$$

be the (closed, convex, simplicial, full in its $(k+1)$-dimensional span) normal cone of G at p, and for a unit vector u in the relative interior of $N(G, p)$ define the directional cap

$$C_p^u(h) \;:=\; \{x \in G :\; \langle x, u\rangle \;\ge\; \langle p, u\rangle - h\}.$$

Proposition 1.3 (Universal cap exponent)**.** *Let (K, h_k) be admissible and $p \in E$. For every unit u in the relative interior of $N(G, p)$ there are $0 < \gamma_1 \le \gamma_2$ and $h_* > 0$, depending on n, k, θ_0, the curvature band, and the angular distance of u to $\partial N(G, p)$, such that*

$$\gamma_1\, h^{\frac{n+1+k}{2}} \;\le\; \operatorname{vol}_n\big(C_p^u(h)\big) \;\le\; \gamma_2\, h^{\frac{n+1+k}{2}} \qquad (0 < h \le h_*), \qquad \frac{n+1+k}{2} = \frac{2n-j}{2}.$$

The exponent is independent of the angles θ_0 and of u within the cone.

Theorem 1.4 (Section lemma, general position)**.** *Let (K, h_k) be admissible. There are $a > 0$ and $h_0 > 0$, depending only on n, k, θ_0 and the curvature band, such that for every unit vector u and every $0 < h \le h_0$, writing $\tau_{\max} := \max_{x \in G}\langle x, u\rangle$ and $A_u(h) := \operatorname{vol}_{n-1}(G \cap \{\langle x, u\rangle = \tau_{\max} - h\})$,*

$$A_u(h) \;\ge\; a\, h^{\frac{n-1+k}{2}}.$$

Consequently the UDC for the truncation set h_k holds, with probability at least $1 - \frac{1}{N}$ for $N \ge N_0(n, k, \theta_0, K)$, at scaling factor

$$\ell_N \;=\; C\left(\frac{\log N}{N}\right)^{2/(n+1+k)}.$$

Theorem 1.5 (Lower bound, general position)**.** *Let (K, h_k) be admissible. There is $c > 0$ such that with probability $1 - o(1)$ the doubling inequality fails for some slab truncated by h_k at every grid scale below $c\,(\log N/N)^{2/(n+1+k)}$; hence the UDC requires $\ell \geq c\,(\log N/N)^{2/(n+1+k)}$. The only configuration-dependent input is the two-sided cap bound*

$$c_1\, h^{\frac{n+1+k}{2}} \;\leq\; \mu\big(C_p(h)\big) \;\leq\; c_2\, h^{\frac{n+1+k}{2}}, \qquad c_1 = \omega(N_p)\cdot\kappa_{\mathrm{geom}} > 0,$$

where $\omega(N_p)$ is the normalized solid angle of the normal cone $N(G,p)$ and $\kappa_{\mathrm{geom}} > 0$ collects the curvature and conjugate-frame constants.

Corollary 1.6 (The interpolation law in general position)**.** *Let K be a C^2 strictly convex body with curvatures in $[\lambda_{\min}, \lambda_{\max}]$ and let h_k be a general-position simplicial truncation with dihedral angles bounded below by θ_0, whose lowest stratum E has dimension $j = n-1-k$. Then the critical scaling of the uniform doubling condition is*

$$\ell_N \;=\; \Theta\Big(\big(\log N/N\big)^{2/(2n-j)}\Big), \qquad 1 \leq j \leq n-1,$$

with exponent and logarithm exact and all constants depending only on n, k, θ_0 and the curvature band. For $j = 0$ (simplicial cones, $k = n-1$) the conical scaling $\Theta((\log N/N)^{1/n})$ of [3, 4] applies. In particular the law holds for every simplicial cone and for every nonsingular affine image of the orthogonal model.

Corollary 1.6 is the general-position completion of [7, Theorem 1.1]: it removes the roundness of the ball and the orthogonality of the cuts, the only two special features of the model, while keeping both halves of the bound and the constant in the logarithm. The route that does *not* work — straightening the configuration to the model by a linear change of variables — is discussed in §6; the obstruction there is what forces the intrinsic conjugate-frame argument of §§2–5.

Organization. Section 2 sets up depth coordinates at the stratum and proves the universal exponent (Proposition 1.3) and the two-sided cap bound (Lemma 2.3). Section 3 proves Theorem 1.5. Section 4 proves the section lemma (Theorem 1.4) in general position, by induction on k through the slicing closure of the admissible class. Section 5 assembles the slab bound and the upper half of Theorem 1.4. Section 6 records the John-position obstruction and the residual open case.

2 Depth coordinates and the cap exponent

2.1 Floor coordinates at a stratum point

Fix $p \in E$ and a unit vector u in the relative interior of $N(G,p)$. We build coordinates adapted to u in which the convex floor of G near p is explicit.

Let $T := T_p\partial K = \nu_K(p)^{\perp}$, split as $T = \Pi V \oplus T_pE$ with $\Pi V = \mathrm{span}\{\bar a_1, \dots, \bar a_k\}$ (dimension k) and $T_pE = \{v \in T : \langle v, a_i\rangle = 0\}$ (dimension $m := j = n-1-k$). Write $u = u^{\nu}\,\nu_K(p) + u^V + u^E$ with $u^{\nu} = \langle u, \nu_K(p)\rangle$, $u^V \in \Pi V$, $u^E \in T_pE$. Because $u \in \mathrm{relint}\, N(G,p)$ and $N(G,p) = \mathrm{cone}\{\nu_K(p), a_1, \dots, a_k\}$, the expansion $u = \beta_0\nu_K(p) + \sum_i \beta_i a_i$ has all $\beta_i > 0$; since u^E is the component of u orthogonal to $\nu_K(p)$ and to every a_i, and the a_i span $V \supseteq \Pi V$, one checks $u^E = 0$. Thus u lies in the $(k+1)$-plane $\mathrm{span}\{\nu_K(p)\} \oplus \Pi V$, the span of the cone, as it must.

Choose orthonormal coordinates on $\mathbb{R}^n$ as follows. The first coordinate is $z := \langle p-x, \nu_K(p)\rangle \geq 0$, the depth below the tangent plane at p (nonnegative on K near p by convexity). On T_pE choose an orthonormal basis $\eta_1, \dots, \eta_m$ and set $y_a := \langle x-p, \eta_a\rangle$. On ΠV we use the conjugate coordinates $s_i := \langle x-p, a_i\rangle$ of Definition 1.2; by (1) the change of basis between $(s_1, \dots, s_k)$ and an orthonormal basis of ΠV has condition number bounded by $g_2/g_1 =: \Gamma(\theta_0, k)$, and we will only ever need this two-sided comparison. Thus $(z, y, s) \in \mathbb{R} \times \mathbb{R}^m \times \mathbb{R}^k$ is a system of *bounded-distortion* coordinates centered at p, with Jacobian bounded above and below by constants depending on θ_0, k.

Lemma 2.1 (Convex floor)**.** *In the coordinates (z, y, s) there are $r, L > 0$ depending on n, k, θ_0 and the curvature band such that for $|y|, |s| \leq r$, the boundary ∂K is the graph*

$$z \;=\; \Phi_0(y,s) \;:=\; \tfrac{1}{2}\,Q(y,s) + O\big((|y|+|s|)^3\big), \qquad \lambda_{\min}\,(|y|^2+|s|^2) \;\leq\; Q(y,s) \;\leq\; \lambda_{\max}\,(|y|^2+|s|^2),$$

where Q is the second fundamental form II_p read in these coordinates, and the O-term has C^0 norm $\leq L(|y|+|s|)^3$. Moreover $G \cap \{|y|, |s| \leq r\} = \{z \geq \Phi_0(y,s)\} \cap \{s_1 \leq 0, \dots, s_k \leq 0\}$ within $\{|y|, |s| \leq r\}$.

Proof. Since $\nu_K(p)$ is the outer normal, ∂K is locally the graph of a C^2 function $z = \phi(y, s)$ over T with $\phi(0) = 0$, $\nabla\phi(0) = 0$, and Hessian II_p, whose eigenvalues are the principal curvatures in $[\lambda_{\min}, \lambda_{\max}]$. Taylor expansion gives Φ_0 with the stated quadratic bounds and a C^2 remainder, uniformly because the curvatures are bounded and K is C^2 with a modulus controlled on the compact stratum. The constraint description holds because $\langle x-p, a_i\rangle = s_i$ exactly and the curved faces $\{\langle x, a_i\rangle = c_i\}$ do not enter the *interior* normal stratum within the small box (they bound G only where some $s_i = 0$, already included). □

We separate the directions of u by whether the floor is flat or curved along them. By the previous paragraph $u = u^\nu \nu_K(p) + u^V$ with $u^\nu > 0$ (else u is tangent and not interior to the cone). Write the slab depth $\langle p - x, u\rangle$, i.e. the cap function, in coordinates. Using $x - p = -z\,\nu_K(p) + \sum_a y_a \eta_a + \Pi(x-p)$ and that $u^E = 0$,

$$\begin{aligned} \langle p-x, u\rangle &= u^\nu z - \langle x-p, u^V\rangle \\ &= u^\nu z - \sum_{i=1}^{k} \beta_i\,\langle x-p, a_i\rangle \;=\; u^\nu z - \sum_{i=1}^{k} \beta_i\, s_i, \end{aligned} \tag{2}$$

because $u^V = \sum_i \beta_i \bar{a}_i$ pairs with $x-p$ through $\langle \cdot, a_i\rangle = s_i$ on T (the $\nu_K(p)$-component of $x-p$ is orthogonal to ΠV). Here $\beta_i > 0$ are the cone coefficients of u, and by (GP2) and (1) they obey $\beta_i \in [\beta_-, \beta_+]$ with $\beta_- > 0$ depending on the angular distance of u to $\partial N(G, p)$.

2.2 Proof of Proposition 1.3

Proof. On $C_p^u(h)$ we have $\langle p-x, u\rangle \leq h$, $s_i \leq 0$, and $z \geq \Phi_0(y,s)$. Substitute (2): with $\sigma_i := -s_i \geq 0$,

$$C_p^u(h) \cap \{|y|, |s| \leq r\} = \Big\{(y, \sigma, z) :\; \sigma_i \geq 0,\; z \geq \Phi_0(y, -\sigma),\; u^\nu z + \sum_i \beta_i \sigma_i \leq h\Big\}. \tag{3}$$

The cap is contained in this box for $h \leq h_*$ small, because the cap radius is $O(\sqrt{h})$ (shown below) and then $\Phi_0 \geq \lambda_{\min}(|y|^2 + |s|^2)/2$ forces $|y|, |s| \leq r$.

Rescaling. Put $y = \sqrt{h}\,Y$, $\sigma = h\,S$ (note: linear in h for the truncated directions, square-root for the tangent-to-stratum directions), and $z = h\,Z$. Then $dy\,d\sigma\,dz = h^{m/2} \cdot h^k \cdot h\ dY\,dS\,dZ$, and

$$\Phi_0(y, -\sigma) = \tfrac{1}{2}Q(\sqrt{h}Y, -hS) + O(h^{3/2}) = \tfrac{h}{2}Q(Y, 0) + O(h^{3/2}),$$

because $Q(\sqrt{h}Y, -hS) = h\,Q(Y, 0) - h^{3/2}(\text{cross}) + h^2 Q(0, S)$ and the last two terms are $o(h)$. Hence after dividing the cap inequality by h,

$$C_p^u(h) \iff \Big\{(Y, S, Z) :\ S_i \geq 0,\ Z \geq \tfrac{1}{2}Q(Y, 0) + o(1),\ u^\nu Z + \sum_i \beta_i S_i \leq 1 + o(1)\Big\}. \quad (4)$$

The limiting region (as $h \to 0$) is

$$R := \Big\{(Y, S, Z) :\ S \geq 0,\ Z \geq \tfrac{1}{2}Q(Y, 0),\ u^\nu Z + \textstyle\sum_i \beta_i S_i \leq 1\Big\},$$

a fixed set of finite, positive n-volume: finiteness because $Q(Y, 0) \geq \lambda_{\min}|Y|^2$ confines Y to $|Y| \leq \sqrt{2/(u^\nu \lambda_{\min})}$ and then $Z \leq 1/u^\nu$, $S_i \leq 1/\beta_i$; positivity because the relative interior is nonempty (take $Y = 0$, small $S_i > 0$, Z slightly above 0). The $o(1)$ perturbations are uniform on compact sets and the regions are nested up to $1 \pm o(1)$ dilations, so by dominated convergence

$$\mathrm{vol}_n\big(C_p^u(h)\big) = h^{\,1+k+m/2}\big(\mathrm{vol}_n(R) + o(1)\big), \qquad 1 + k + \tfrac{m}{2} = \frac{(n-1) + 1 + k}{2} = \frac{n + 1 + k}{2}.$$

The exponent counts 1 for the depth z (length h), 1 for each truncated direction s_i (length h, a *linear* floor $\beta_i\sigma_i$), and $\frac{1}{2}$ for each of the $m = j$ stratum directions y_a (length $\sqrt{h}$, a *quadratic* floor $\frac{1}{2}Q$). None of these scalings sees the angles, only the dichotomy linear-vs-quadratic floor; this proves the angle-independence and gives γ_1, γ_2 as $\mathrm{vol}_n(R)(1 \mp o(1))$. Finally the cap-radius claim: on R, $|Y|, |S|, |Z| = O(1)$, so on $C_p^u(h)$, $|y| = O(\sqrt{h})$, $|s| = O(h)$, $z = O(h)$, hence $|x - p| = O(\sqrt{h})$, justifying the box confinement for $h \leq h_*$. □

Remark 2.2. Proposition 1.3 is the conceptual core. In the orthogonal model the same computation has $a_i = e_i$, β_i the positive coordinates of u, and $Q = \mathrm{Id}$ (the round sphere), so the floor is $\frac{1}{2}|Y|^2 + \sum \beta_i\sigma_i$ and one reads off $(n + 1 + k)/2$ directly. General position changes Q to an anisotropic but uniformly elliptic form and changes the truncated directions from coordinate axes e_i to the conjugate directions f_i; neither change touches the count of linear versus quadratic floor directions, which is all the exponent records. This is why no rounding of K or orthogonalization of the a_i is needed for the exponent — and, as §6 shows, no such rounding exists.

2.3 The stratum cap and its mass

For the lower bound we need a canonical cap at each $p \in E$ and its mass with the correct constant. Let

$$u_p := \frac{\nu_K(p) + \sum_{i=1}^k a_i}{\big|\nu_K(p) + \sum_i a_i\big|}, \qquad C_p(h) := C_p^{u_p}(h) = \{x \in G :\ \langle x, u_p\rangle \geq \langle p, u_p\rangle - h\}.$$

By (GP1)–(GP2) the vector $\nu_K(p) + \sum_i a_i$ is a fixed positive combination of the cone generators, hence $u_p \in \mathrm{relint}\, N(G, p)$ with angular distance to $\partial N(G, p)$ bounded below by a function of θ_0, k; $\langle p, u_p\rangle = \max_{x \in G}\langle x, u_p\rangle$ because $u_p \in N(G, p)$ (supporting at p). So $C_p(h)$ is a cap of the type in Proposition 1.3, uniformly over E.

Lemma 2.3 (Stratum cap mass and footprint)**.** *There are $0 < c_1 \le c_2$ and $h_1 > 0$ depending on n, k, θ_0 and the curvature band, and $c_3 = c_3(\theta_0, k)$, such that for all $p \in E$ and $0 < h \le h_1$,*

$$c_1\, h^{\frac{n+1+k}{2}} \;\le\; \mu\big(C_p(h)\big) \;\le\; c_2\, h^{\frac{n+1+k}{2}}, \qquad C_p(h) \subseteq \{x :\ |x-p| \le c_3\sqrt{h}\}.$$

Moreover one may take $c_1 = \omega(N_p) \cdot \kappa_{\mathrm{geom}}$, where $\omega(N_p) \in (0,1]$ is the normalized solid angle of $N(G,p)$ and $\kappa_{\mathrm{geom}} > 0$ depends only on the curvature band and θ_0, k.

Proof. Apply Proposition 1.3 with $u = u_p$: it gives $\mathrm{vol}_n(C_p(h)) = h^{(n+1+k)/2}(\mathrm{vol}_n(R_p) + o(1))$ and the footprint $|x-p| \le c_3\sqrt{h}$, and dividing by $\mathrm{vol}_n(K)$ gives the measure bound with $c_2 = 2\,\mathrm{vol}_n(R_p)/\mathrm{vol}_n(K)$, $c_1 = \frac{1}{2}\,\mathrm{vol}_n(R_p)/\mathrm{vol}_n(K)$ for $h \le h_1$, the constants uniform in p because R_p depends continuously on the compact data $(\nu_K(p), a_i, \mathrm{II}_p)$ and is bounded away from degenerate.

For the solid-angle form of c_1, compute $\mathrm{vol}_n(R_p)$ from below by the cone factorization. In (3), drop the curvature in Z (i.e. restrict to $Z \ge 0$ but allow $Z \ge \frac{1}{2}Q(Y,0)$); the region R_p contains the product slab

$$\{Y : \tfrac{1}{2}Q(Y,0) \le \tfrac{1}{2u^\nu}\} \times \{(S,Z) :\ S \ge 0,\ Z \ge 0,\ u^\nu Z + \textstyle\sum_i \beta_i S_i \le \frac{1}{2}\},$$

whose Y-factor has volume $\ge \det(\mathrm{II}_p)^{-1/2}\ \mathrm{vol}_m(\{|\cdot| \le \rho\}) \ge \lambda_{\max}^{-m/2}\,\omega_m \rho^m$ (an ellipsoid with semi-axes $\ge (u^\nu \lambda_{\max})^{-1/2}$), and whose (S,Z)-factor is the simplex $\{S \ge 0, Z \ge 0,\ u^\nu Z + \sum \beta_i S_i \le \frac{1}{2}\}$ of volume $\frac{1}{(k+1)!}\,(\frac{1}{2})^{k+1}/(u^\nu \prod_i \beta_i)$. Now $u^\nu \prod_i \beta_i$ is, up to the bounded Gram determinant of the cone generators, exactly the reciprocal of the normalized solid angle of $N(G,p) = \mathrm{cone}\{\nu_K(p), a_i\}$ at unit radius: indeed the simplex $\{\beta_0 \ge 0, \beta_i \ge 0,\ \beta_0 u^\nu + \sum \beta_i \beta_i \le \frac{1}{2}\}$ in cone-coordinates has Euclidean volume proportional to $\sqrt{\det \mathrm{Gram}(\nu_K(p), a_i)}$, and that determinant is bounded in $[g_0^{(k+1)/2}, (k+1)^{(k+1)/2}]$ by (GP2). Collecting the bounded curvature and Gram factors into κ_{geom} yields $\mathrm{vol}_n(R_p) \ge \omega(N_p)\,\kappa_{\mathrm{geom}}\,\mathrm{vol}_n(K)$, hence $c_1 = \omega(N_p)\kappa_{\mathrm{geom}} > 0$. Positivity of $\omega(N_p)$ is exactly (GP1): a degenerate (lower-dimensional, $\omega = 0$) cone would mean $\nu_K(p), a_i$ linearly dependent. □

Remark 2.4. This is where the orthogonal model's exact fraction 2^{-k} is replaced. There, $N(G,p) = \mathrm{cone}\{p, e_1, \dots, e_k\}$ is a right-angled corner, its normalized solid angle is exactly 2^{-k} of the half-space solid angle, and the all-negative orthant keeps that fraction by reflection symmetry ([7, Lemma 2.2]). In general position the corner is oblique, the reflection symmetry is gone, and the surviving fraction of the cap is the genuine normalized solid angle $\omega(N_p)$ of the oblique cone — still a fixed positive number by (GP2), but no longer a clean power of two. Only positivity, not the exact value, is needed downstream.

Lemma 2.5 (Stall)**.** *Fix $\kappa > 1$, $\sigma > 0$, $\delta \ge \frac{3}{N}$, $p \in E$ and $0 < h \le \frac{H}{4}$, $H \le h_1$. If exactly one sample point lies in $C_p(H)$ and it lies in $C_p(h)$, then for every $w \le H/\kappa$ the slab with direction u_p, center that sample point, and truncation set h_k violates the doubling inequality; hence the UDC fails for every $\ell \le H/\kappa^2$.*

Proof. Verbatim the companion stall lemma [7, Lemma 2.4]: for $w' \le H$ the slab is contained in $C_p(h + \frac{w'}{2}) \subseteq C_p(H)$, so the empirical masses of S_w and $S_{\kappa w}$ both equal $\frac{1}{N} < \min\{\delta, (1+\sigma)\frac{1}{N}\}$. Nothing in the argument uses orthogonality or roundness; it uses only that caps are nested and that $C_p(H)$ holds a single point. □

3 The lower bound

Proof of Theorem 1.5. This is the second-moment scheme of [6, 7] verbatim, fed by Lemma 2.3. Set $q := (4\kappa^2)^{(n+1+k)/2}\, c_2/c_1$, $\varepsilon_3 := \frac{j}{4q(2n-j)}$, $\tau_N := \varepsilon_3 \log N$. Define h_N by $\mu(C_p(h_N)) = \tau_N/N$; this is well defined and, by Lemma 2.3, $h_N \asymp (\tau_N/N)^{2/(n+1+k)}$ uniformly in p (the constants c_1, c_2 are p-independent). Put $H_N := 4\kappa^2 h_N$, so $\mu(C_p(H_N)) \le q\,\tau_N/N$.

By the footprint bound the caps have radius $\le c_3\sqrt{H_N}$ around their centers $p \in E$. Since E is a compact C^2 j-manifold, a maximal $2c_3\sqrt{H_N}$-separated family $p_1, \dots, p_M \in E$ has

$$M \;\ge\; c\,\big(\sqrt{H_N}\big)^{-j} \;\ge\; c'\,(N/\tau_N)^{\frac{j}{n+1+k}}, \qquad \frac{j}{n+1+k} = \frac{j}{2n-j},$$

because the j-dimensional volume of E is a fixed positive constant and a metric ball of radius ρ in E has volume $\asymp \rho^j$ (curvatures of E are bounded, by admissibility). The outer caps $C_{p_\alpha}(H_N)$ are pairwise disjoint: if $x \in C_{p_\alpha}(H_N) \cap C_{p_\beta}(H_N)$ then $|x - p_\alpha|, |x - p_\beta| \le c_3\sqrt{H_N}$, so $|p_\alpha - p_\beta| \le 2c_3\sqrt{H_N}$, contradicting separation.

Let E_α be the event that $C_{p_\alpha}(H_N)$ contains exactly one sample point and that point lies in $C_{p_\alpha}(h_N)$, and $X := \sum_\alpha \mathbf{1}_{E_\alpha}$. Exactly as in the companion,

$$\mathbb{P}(E_\alpha) = N\,\mu(C_{p_\alpha}(h_N))\,(1 - \mu(C_{p_\alpha}(H_N)))^{N-1} \ge \tau_N\, N^{-2q\varepsilon_3},$$

$$\mathbb{E}X = \sum_\alpha \mathbb{P}(E_\alpha) \ge c'\, N^{\frac{j}{2n-j} - 2q\varepsilon_3}\, \tau_N^{\,1-\frac{j}{2n-j}} \;\ge\; c'\, N^{\frac{j}{2(2n-j)}} \to \infty,$$

using $2q\varepsilon_3 = \frac{j}{2(2n-j)}$. The outer caps being disjoint, the events E_α are negatively associated up to the standard correction, and the variance ratio is $\exp(O(N\,\mu(C(H_N))^2)) = 1 + o(1)$; Chebyshev gives $\mathbb{P}(X = 0) \le (\mathbb{E}X)^{-1} + o(1) = o(1)$. On $\{X \ge 1\}$, Lemma 2.5 forces $\ell > H_N/\kappa^2 = 4h_N \ge c\,(\log N/N)^{2/(n+1+k)}$. □

4 The section lemma in general position

We prove Theorem 1.4. As in [7] it is equivalent, and more convenient, to prove the cap-mass lower bound

$$\Lambda_u(h) := \mu\big(\{x \in G :\ \langle x, u\rangle \ge \tau_{\max} - h\}\big) \;\ge\; b\,h^{\frac{n+1+k}{2}} \qquad (0 < h \le h_0) \tag{5}$$

for every unit u, together with a bulk anchor; the conversion to sections is identical to the model and is recalled at the end. The proof is by induction on the number k of truncations. The induction is legitimate because the admissible class is closed under the slicing used below (Lemma 4.1).

4.1 Closure under slicing

Lemma 4.1 (Slices of admissible configurations are admissible)**.** *Let (K, h_k) be admissible with $k \ge 1$, fix $i_0 \in \{1, \dots, k\}$ and a level $s \in \mathbb{R}$ near c_{i_0}, and set*

$$K_s \;:=\; K \cap \{\langle x, a_{i_0}\rangle = s\}, \qquad h_{k-1}^{(s)} \;:=\; \{(a_i, c_i) : i \ne i_0\}\big|_{K_s}.$$

For s in a one-sided neighborhood of c_{i_0} on which $K_s \ne \emptyset$, the pair $(K_s, h_{k-1}^{(s)})$ is an admissible $(n-1, k-1)$ configuration with the same angular constant $\theta_0' \ge \theta_0/2$ and curvature band

$[\lambda_{\min}, \lambda'_{\max}]$, $\lambda'_{\max} \le \lambda_{\max}/\sin\theta_0$, *and its stratum* $E_s = E \cap \{\langle x, a_{i_0}\rangle = s\}$ *has dimension* j *(one less ambient dimension, one fewer constraint).*

Proof. K_s is the intersection of the C^2 strictly convex body K with an affine hyperplane $\{\langle \cdot, a_{i_0}\rangle = s\}$ transversal to ∂K near E (transversality is (GP1): a_{i_0} is independent of ν_K along E, so the hyperplane is not tangent). A transversal hyperplane section of a C^2 strictly convex body is a C^2 strictly convex body in the hyperplane, with principal curvatures bounded below by $\lambda_{\min}$ and above by $\lambda_{\max}/\sin\theta(a_{i_0}, \nu_K) \le \lambda_{\max}/\sin\theta_0$ (the section curvature is the normal curvature divided by the sine of the incidence angle). The remaining normals a_i ($i \ne i_0$), projected into the hyperplane $a_{i_0}^{\perp}$, are the truncating normals of the slice; their mutual angles and their angles to the section normal stay in $[\theta_0/2, \pi - \theta_0/2]$ because projecting along a_{i_0} (which makes angle $\ge \theta_0$ with each of them and with ν_K) distorts angles by at most $\theta_0/2$ for s close enough to c_{i_0}. Linear independence (GP1) of $\{\nu_{K_s}, a_i : i \ne i_0\}$ in the hyperplane follows from (GP1) for the full list minus a_{i_0}. The stratum identity is immediate. □

Remark 4.2. Lemma 4.1 is the general-position analogue of the fact, used freely in [7], that a coordinate slice of the orthogonal model $B^n \cap \{x_i \le 0\}$ is the orthogonal model $B^{n-1} \cap \{x_i \le 0\}$ in one lower dimension. There the slice is again a round ball cut by coordinate half-spaces; here it is an oblique section, a C^2 convex body cut by oblique half-spaces, but still admissible. This closure is what lets the induction run without ever leaving the class.

4.2 Proof of the cap-mass bound (5)

Proof. We induct on k. The base case $k = 0$ is the smooth strictly convex body: by Lemma 2.1 with no constraints, the cap of depth h in direction $u = \nu_K(p_u)$ (with p_u the unique boundary maximizer of $\langle \cdot, u\rangle$) has, after the rescaling $y = \sqrt{h}Y$, $z = hZ$, $\mu(C(h)) \ge c(n, \lambda_{\max})h^{(n+1)/2}$, the standard cap bound for a body with bounded curvature [3]. Assume (5) for all admissible $(n-1, k-1)$ configurations.

Fix an admissible (K, h_k) and a unit u. Let $p_u \in \partial K$ maximize $\langle \cdot, u\rangle$ over G; by convex duality p_u lies in the stratum whose normal cone contains u. Let $S \subseteq \{1, \dots, k\}$ be the active set at p_u, i.e. the constraints with $\langle p_u, a_i\rangle = c_i$. We distinguish three cases by the position of u relative to the normal cone $N(G, p_u) = \mathrm{cone}\{\nu_K(p_u), a_i : i \in S\}$.

Case 1 (u interior to the full normal cone; $S = \{1, \dots, k\}$ and $u \in \mathrm{relint}\, N(G, p_u)$). Then $p_u \in E$ and Proposition 1.3 (via Lemma 2.3's proof, which only used interiority and admissibility) gives directly

$$\Lambda_u(h) = \mu(C^u_{p_u}(h)) \;\ge\; \gamma_1\, h^{(n+1+k)/2},$$

with $\gamma_1 > 0$ depending on the angular distance of u to $\partial N(G, p_u)$, hence on θ_0, k uniformly. This is (5).

Case 2 (some constraints inactive at p_u; $S \subsetneq \{1, \dots, k\}$). Then near p_u the constraints $\{i \notin S\}$ are slack: $\langle p_u, a_i\rangle < c_i$ for $i \notin S$, so there is $\rho > 0$ with $\langle x, a_i\rangle < c_i$ for all $i \notin S$ and all x with $|x - p_u| \le \rho$. For $h \le h_0(\rho)$ the cap $C^u(h)$ has radius $\le c_3\sqrt{h} \le \rho$ (footprint bound), hence lies in $\{x : \langle x, a_i\rangle < c_i,\ i \notin S\}$, where G agrees with the sub-configuration $G_S := K \cap \bigcap_{i \in S}\{\langle x, a_i\rangle \le c_i\}$. Now $(K, h_{|S|})$ with truncation set $\{(a_i, c_i) : i \in S\}$ is an admissible $(n, |S|)$ configuration with $|S| \le k$ active constraints and the *same* θ_0 (a sub-list of admissible normals is admissible), and $u \in \mathrm{relint}\, N(G_S, p_u)$ with p_u on *its* lowest stratum $E_S \supseteq E$. If $|S| < k$ apply the induction hypothesis in the form already proved for fewer constraints (Case

1 for G_S when u is interior to $N(G_S, p_u)$, else recurse into Case 2/3 for G_S); the exponent it yields is $(n+1+|S|)/2 \le (n+1+k)/2$, and since $h \le 1$,

$$\Lambda_u(h) = \mu(C^u(h) \cap G_S) \;\ge\; b_S\, h^{(n+1+|S|)/2} \;\ge\; b_S\, h^{(n+1+k)/2}.$$

The bound $h \le 1$ only helps; we lose nothing because the larger exponent is the worst case and it is achieved exactly when all k constraints are active, i.e. in Cases 1 and 3.

Case 3 (u near-tangent to an active facet; $S = \{1, \dots, k\}$ but $u \in \partial N(G, p_u)$ or just inside, near the wall $\{\beta_{i_0} \to 0\}$). This is the inductive heart. Suppose u is within angle $\theta_0/4$ of the facet of $N(G, p_u)$ opposite the generator a_{i_0}, i.e. the cone coefficient β_{i_0} of u is the smallest and may be near 0. We slice out a_{i_0}. For $s \in [\, c_{i_0} - \frac{h}{2|u^{(i_0)}|},\, c_{i_0}]$ — an interval of length $\frac{h}{2|u^{(i_0)}|} \ge \frac{h}{2}$, where $u^{(i_0)} := \langle u, a_{i_0}\rangle$ is bounded below by a constant times β_{i_0} but the interval length uses only $|u^{(i_0)}| \le 1$ — consider the slice

$$K_s = K \cap \{\langle x, a_{i_0}\rangle = s\}, \qquad G_s = G \cap \{\langle x, a_{i_0}\rangle = s\} = K_s \cap \bigcap_{i \ne i_0} \{\langle x, a_i\rangle \le c_i\}.$$

By Lemma 4.1, $(K_s, h^{(s)}_{k-1})$ is admissible of type $(n-1, k-1)$. On the slice, write $\langle x, u\rangle = u^{(i_0)} s + \langle x, u_\perp\rangle$, where $u_\perp$ is the orthogonal projection of u onto $a_{i_0}^\perp$, $|u_\perp| \le 1$. The unit direction $u_\perp/|u_\perp|$ is admissible for the slice (it is some unit vector in the hyperplane), and the maximizer of $\langle \cdot, u_\perp\rangle$ over G_s approaches p_u as $s \to c_{i_0}$; the maximum value $\tau^{(s)}_{\max}$ differs from $\tau_{\max} - u^{(i_0)} s$ by at most $O(h)$ for s within $O(h)$ of c_{i_0}, by the C^2 dependence of the section on s (curvatures bounded via Lemma 4.1). Hence the cap of depth h for (G, u) contains, over each such s, the image of the cap of depth $h_* := h/(4|u_\perp|) \ge h/4$ for $(G_s, u_\perp/|u_\perp|)$, once we budget the $\le 2h$ of depth spent on the s-displacement and the section drift within a total depth $4h$ (and then replace h by $h/4$ throughout, which only changes constants). By the induction hypothesis on $(n-1, k-1)$ at depth $\min(h_*, h_0)$ — if $h_* > h_0$ use monotonicity $\Lambda(h_*) \ge \Lambda(h_0) \ge b\, h_0^{(n+k-1)/2} \ge b' h^{(n+k-1)/2}$ — the slice cap has $(n-1)$-mass $\ge b(n-1, k-1)\, h^{((n-1)+1+(k-1))/2} = b\, h^{(n+k-1)/2}$. Integrating over the s-interval of length $\ge h/2$ (the co-area factor $|\nabla\langle \cdot, a_{i_0}\rangle| = 1$ is constant) by Fubini,

$$\Lambda_u(h) \;\ge\; \frac{h}{2} \cdot b(n-1, k-1)\, h^{\frac{n+k-1}{2}} \;=\; b'\, h^{1+\frac{n+k-1}{2}} \;=\; b'\, h^{\frac{n+1+k}{2}}.$$

This proves (5) in Case 3 and closes the induction.

The three cases are exhaustive: either all k constraints are active at p_u (Cases 1 and 3, split by whether u is interior or near-tangent to the cone) or some are inactive (Case 2). In every case $\Lambda_u(h) \ge b\, h^{(n+1+k)/2}$ with $b = b(n, k, \theta_0, \lambda_{\min}, \lambda_{\max}) > 0$, proving (5). □

4.3 Conversion to sections

Exactly as in [7, §4]. By Brunn–Minkowski the section area $A_u(\cdot)$ is unimodal in the depth. The thickness ball $B(c^-, r_0) \subseteq G$ provides a bulk depth $h^\dagger \ge c(\theta_0, k) > 0$ at which $A_u(h^\dagger) \ge a_1 > 0$ (its $\langle \cdot, u\rangle$-range is an interval of length $2r_0$ inside the range of G, with central sections of radius $\ge \frac{\sqrt{3}}{2} r_0$). For $h \le h_0 \le h^\dagger$: below the mode of A_u, A_u is nondecreasing on $[0, h]$ and $A_u(h) \ge \Lambda_u(h)/h \ge b\, h^{(n-1+k)/2}$ by (5); above the mode, unimodality gives $A_u(h) \ge A_u(h^\dagger) \ge a_1 \ge a_1 h^{(n-1+k)/2}$. Either way $A_u(h) \ge a\, h^{(n-1+k)/2}$, which is Theorem 1.4.

Remark 4.3. The witness in Case 1 is the conjugate-frame box: take Y, S in the limiting region R of Proposition 1.3 and pull back to a box in x whose f_i-edges have length $\asymp h$ (truncated directions) and whose η_a-edges have length $\asymp \sqrt{h}$ (stratum directions), capped by the fibre of length $\asymp h$ in the u-direction. The orthogonal model's box [7, §4] is the special case $f_i = e_i$; the only structural change is that the truncated edges point along the conjugate directions $f_i \in \operatorname{span}\{a_i\}$ rather than the coordinate axes, which is forced because the constraints are diagonalized by $s_i = \langle x - p, a_i \rangle$, not by the standard coordinates. Case 3 is the mechanism by which face strata inherit the law from lower-dimensional admissible models, one truncation per dimension, through the slicing closure of Lemma 4.1.

5 Slab bound and the upper bound

Theorem 5.1 (Slab volume)**.** *There are $c > 0$ and $L_0 > 0$, depending only on n, k, θ_0 and the curvature band, with*

$$\operatorname{vol}_n\Big(G \cap \{|\langle y - x, u\rangle| \le L\}\Big) \;\ge\; c\, L^{\frac{n+1+k}{2}} \qquad (x \in G,\ u \in S^{n-1},\ 0 < L \le L_0).$$

Proof. Anchor–unimodality as in [3, 7]. Apply Theorem 1.4 to u and to $-u$: there are anchors at depth $L/4$ from each u-extreme of G with section value $\ge a\,(L/4)^{(n-1+k)/2}$. By Brunn–Minkowski unimodality the section area exceeds the smaller anchor value throughout the middle of the $\langle \cdot, u\rangle$-range, whose complement has length $L/2$; the range has length $\ge 2r_0 \ge 2L$ by thickness; and the slab keeps length $\ge L$ inside the range, of which $\ge L/2$ is in the middle. Integrating the section area over that middle length gives $\ge \frac{L}{2} \cdot a(L/4)^{(n-1+k)/2} = c\, L^{(n+1+k)/2}$. □

Proof of Theorem 1.4, upper bound. Insert Theorem 5.1 into the high-probability skeleton of [2, 3], identical to [7, §5]. The population dichotomy holds with thickness radius r_0; the VC relative-deviation bounds apply to slabs cut by the k half-spaces of h_k (a slab truncated by h_k is an intersection of $k+2$ half-spaces, VC dimension $O(nk)$, so the uniform deviation $u_N \asymp \log N/N$ controls all of them simultaneously); and a sample point in S_w yields $\mu(S_{\kappa w}) \ge 32 u_N$ for every $w \ge \ell_N := C\,(u_N)^{2/(n+1+k)} \asymp (\log N/N)^{2/(n+1+k)}$, by Theorem 5.1 with $L = \kappa w$. This proves the UDC at scaling ℓ_N with probability $\ge 1 - \frac{1}{N}$. □

Theorems 1.4 and 1.5 together give Corollary 1.6.

6 Why rounding fails, and the open case

6.1 The John-position obstruction

The shortest conceivable proof of Corollary 1.6 would map the admissible configuration to the orthogonal model by a single affine change of variables and quote [7]. This is provably impossible already at the linear level, and the obstruction is instructive.

Proposition 6.1 (No simultaneous rounding and orthogonalization)**.** *Let $a_1, \dots, a_k \in S^{n-1}$ ($k \ge 2$) be unit normals, not pairwise orthogonal. There is no invertible linear map $A \in GL_n(\mathbb{R})$ such that $A(B^n)$ is a ball and $Aa_1, \dots, Aa_k$ are pairwise orthogonal.*

Proof. $A(B^n)$ is a ball iff $A^{\mathsf{T}}A = \lambda I$ for some $\lambda > 0$ (then $A = \sqrt{\lambda}\,O$, O orthogonal). But then for all i, j,

$$\langle Aa_i, Aa_j\rangle = \langle A^{\mathsf{T}}Aa_i, a_j\rangle = \lambda\,\langle a_i, a_j\rangle,$$

so the inner products are scaled by the common factor λ, and the *angles* between the a_i are preserved. In particular $Aa_i \perp Aa_j$ forces $a_i \perp a_j$. If some pair a_i, a_j is non-orthogonal, no such A orthogonalizes it while rounding the ball. □

Thus one cannot have both: any linear map that orthogonalizes the cuts distorts B^n into a genuine ellipsoid (and, for a general K, distorts the curvature band in a direction-dependent way), and any linear map that keeps the body a ball preserves all the cut angles. Affinely, K and the half-spaces are *jointly* rigid up to a similarity: the configuration carries a true modulus, the angles, that no volume-preserving or shape-preserving map can delete. The argument of §§2–5 is therefore intrinsic of necessity: the conjugate frame diagonalizes the constraints *without* touching K, paying for the obliqueness with the anisotropic but uniformly elliptic floor Q of Lemma 2.1 and the solid angle $\omega(N_p)$ of Lemma 2.3, rather than reducing to a right-angled model. This is also why the exponent, which depends only on the linear-versus-quadratic floor dichotomy (Remark 2.2), survives unchanged while the constants acquire θ_0-dependence.

Remark 6.2. For $k = 1$ a single cut can always be orthogonalized trivially, and indeed a similarity reduces one oblique cut of a ball to the model; the obstruction is genuinely a $k \geq 2$ phenomenon, the first place where two angles must coexist. This matches the fact that the orthogonal-model induction of [7] and of §4 only becomes nontrivial at $k \geq 2$.

6.2 Curved truncating surfaces and the companion

Admissibility (Definition 1.1) keeps the truncating boundaries *flat* (half-spaces $\langle x, a_i\rangle \leq c_i$). The curved companion [8] treats the next case, where each flat constraint is replaced by a C^2 truncating surface $g_i(x) \leq 0$, under the same transversality philosophy. For each fixed curved configuration it proves that the exponent survives:

$$\ell_N \asymp \Big(\frac{\log N}{N}\Big)^{2/(2n-j)}.$$

The remaining issue is not the exponent but uniformity of the constants across all curvature-bounded families. With curved truncations the second fundamental forms of the truncating walls enter the same rescaled floor block as the stratum form, so the flat factorization $c_1 = \omega(N_p)\,\kappa_{\mathrm{geom}}$ no longer follows formally. The open problem left to [8] is whether the lower slab/cap constants can be made uniform near multiply-tangent normal directions, or whether one must keep constants depending on the fixed curved configuration.

6.3 Summary of the completed program

Combining the present paper with its companions, for a C^2 strictly convex body cut by half-spaces in general position the critical scaling of the uniform doubling condition is, at every transversal stratum of dimension j,

$$\ell_N \asymp \Big(\frac{\log N}{N}\Big)^{2/(2n-j)} \quad (1 \leq j \leq n-1), \qquad \ell_N \asymp \Big(\frac{\log N}{N}\Big)^{1/n} \quad (j = 0),$$

exponents and logarithms exact, with constants depending only on the dimension, the number of cuts, the dihedral-angle lower bound θ_0, and the curvature band. The orthogonal model

of [7] is recovered as the right-angled, round, flat case $\omega(N_p) = 2^{-k}$, $Q = \mathrm{Id}$, $f_i = e_i$. The finest resolution at which i.i.d. data can certify the doubling condition is again set by the most isolated extremal sample point of the worst boundary stratum of the activation geometry, and the explicit exponent ladder $\frac{2}{n+1}, \frac{2}{n+2}, \ldots, \frac{1}{n}$ governs the transition from smooth to conical truncation regardless of the angles at which the cuts meet. For curved truncating surfaces, [8] settles the fixed-family exponent law and leaves the curvature-band-uniform lower/slab and factorized-constant problems isolated above.

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Declaration on generative AI

During the preparation of this work the author used generative AI tools to accelerate drafting and revision. All mathematical claims, proofs, and bibliographic information were subsequently reviewed and edited by the author, who takes full responsibility for the content of the manuscript.

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